

Equitable AI

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ABSTRACT

Fairness in machine learning (ML) and artificial intelligence (AI) has gained much attention in recent years, which has led to the creation of multiple fairness metrics and frameworks. However, it is still challenging for practitioners to decide which fairness metrics to incorporate in their work and how to effectively incorporate fairness into ML and AI models and systems. ML and AI can benefit from computer-human interaction (CHI) expertise from our backgrounds in empirically grounded research. ML and AI black-box models that are creating biased results are in need of CHI expertise to add the human aspect to the evaluation and creation of AI systems. This will result in what I call Equitable AI systems. In my talk, I will describe an equitable AI system I created to create holistic diversity in higher education admissions, scholarships, hiring, etc. I will also discuss research from a CHI perspective in addressing bias in ML and AI.

KEYWORDS

Equitable AI, Bias in AI, Human-Centered AI

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1 SPEAKER'S BIO

Dr. Juan E. Gilbert is the Andrew Banks Family Preeminence Endowed Professor and Chair of the Computer & Information Science & Engineering Department at the University of Florida where he leads the Human Experience Research Lab. Dr. Gilbert's research involves the creation and evaluation of innovative solutions to real world problems. He integrates a wide variety of people, technology, policy, and cultural lenses to address serious societal issues. His research also examines both HCI and AI through his examinations of Bias in AI, Advanced Learning Technologies, Culturally Aware Computing, and attention to people with a wide variety of disabilities and formative experiences. He is an ACM Fellow, a Fellow of the American Association of the Advancement of Science and a Fellow of the National Academy of Inventors. In 2012, he received the Presidential Award for Excellence in Science, Mathematics, and Engineering Mentoring from President Barack Obama. He also received the American Association for the Advancement of Science

(AAAS) 2014 Mentor Award. In 2018, he received the Computer Research Association's A. Nico Habermann Award. He was also named a 2015-2106 AAAS-Lemelson Invention Ambassador, Speech Technology Luminary by Speech Technology Magazine and a national role model by Minority Access Inc. He is also a National Associate of the National Research Council of the National Academies. Dr. Gilbert was named a Master of Innovation by Black Enterprise Magazine, a Modern-Day Technology Leader by the Black Engineer of the Year Award Conference, the Pioneer of the Year by the National Society of Black Engineers and he received the Black Data Processing Association (BDPA) Epsilon Award for Outstanding Technical Contribution. In 2002, Dr. Gilbert was named one of the nation's top African-American Scholars by *Diverse Issues in Higher Education*. In 2013, the Black Graduate and Professional Student Association at Auburn University named their Distinguished Lecture Series in honor of Dr. Gilbert. He has testified before the Congress on the Bipartisan Electronic Voting Reform Act of 2008 and the U.S. House Committee on Administration for the "2020 Election Security – Perspectives from Voting System Vendors and Experts" In 2006, Dr. Gilbert was honored with a mural painting in New York City by City Year New York, a non-profit organization that unites a diverse group of 17 to 24 year-old young people for a year of full-time, rigorous community service, leadership development, and civic engagement.

Dr. Gilbert's work in Advanced Learning Technologies increases access to technology for those with limited educational opportunities, particularly by providing personalized instruction for those in under-resourced schools that might not otherwise be able to afford this type of engagement. Additionally, in these environments, his work addresses the use of computer and telecommunication technology by diverse populations. Specifically, this research engages culturally relevant learning environments to support education. This ethnocomputing approach improves access to and engagement with education for a wide variety of populations, particularly traditionally underrepresented groups. Likewise, it is well known in higher education research that holistic admissions increases educational opportunities at all levels, but particularly at the undergraduate and graduate school levels. Dr. Gilbert is using his research in AI to understand how clustering algorithms might better process admissions applications in ways that actually serve to increase holistic diversity rather than select for majority dominant groups as traditional algorithmic approaches have done. In pilot studies with several universities, Dr. Gilbert's patented AI algorithm for admissions called Applications Quest has resulted in greater diversity in a fraction of the time it takes the admissions committee while yielding the same academic achievement levels as the committee [1–4]. In short, Dr. Gilbert's research has made substantial impact in areas of educational technology and learning support not only in terms of basic HCI and AI research but also in applying this work

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for great impact to address the systemic barriers to high quality education.

Dr. Gilbert has made a significant impact on broadening the participation of underrepresented groups in computing research specializing in computer-human interaction. He has achieved unprecedented numbers of African-American CS PhDs through his mentorship and leadership in the broadening participation in computing community. He has led multiple NSF projects that have helped broaden the participation of underrepresented groups in computing at the PhD and faculty levels.

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